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# Reading Buys Evidence, Not a Decision: The Cost of Verification and the Limits of Retrieval

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Abstract

Declining to assert an unsupported claim requires verifying it and verifying costs reading. This paper measures what that reading buys on 19,998 human-written claims against an encyclopaedia snapshot, balanced at 6,666 each across supported, refuted and not settled, cost counted in sentences read. From one sentence to a hundred, the share of settleable claims for which an annotator-chosen sentence has been read rises from 0.5892 to 0.9863. Under the stricter definition the shared task's own score uses, a complete annotator set read rather than any one sentence, the same sweep runs lower throughout, from 0.5390 to 0.9453. The same increase takes the share labelled correctly from 0.4001 to 0.4281 against a chance baseline of 0.3333, a paired gain of 0.0280 with a 95 per cent interval of 0.0162 to 0.0396. The two quantities a budget is usually assumed to buy together move by amounts differing fourteenfold. An oracle arm bounds better retrieval: promoting every settleable claim's annotator sentences to the top is worth 0.0253 in accuracy at one sentence and nothing measurable from ten sentences on, 0.0072 averaged over the seven budgets, so the gap is not in finding evidence. Splitting the held-out claims by whether the complete evidence set arrived locates the small accuracy gain away from the evidence: claims whose set never arrives gain 0.0505 between one sentence and a hundred, more than the 0.0418 gained by claims whose set does arrive. The verifier reads retrieval statistics rather than language. That is the measurement rather than a caveat: the reported accuracy is what a system that can locate evidence but not read it achieves. The not-settled decision has the highest recall at every budget and rises most with it: 0.4651 at a hundred against 0.4223 for supported and 0.3985 for refuted. Its precision trails supported's at every budget and refuted's at five of seven, so recall alone does not make it the easiest decision. That evaluation conflates evidence quality with verdict correctness is not new. What this paper adds is its size at an explicit swept reading budget, with an oracle bound on how far better retrieval alone could close it.

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Keywords: claim verification; withholding; retrieval cost; evidence; evaluation methodology; fact checking

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## 1. Introduction

A system asked not to assert what it cannot support has to decide, for each claim, whether the evidence settles it. That decision costs reading: an analyst or a pipeline must look at some number of sentences before deciding. The budget is usually implicit and the result is usually a single accuracy figure.

This paper makes the budget explicit and measures what it buys. The corpus is a set of human-written claims against an encyclopaedia snapshot [1], [2], with each claim labelled supported, refuted or not settled and the sentences a human annotator used recorded only for claims labelled supported or refuted. That evaluation conflates evidence quality with verdict correctness is already argued elsewhere [3]; what is measured here is the size of the gap, in sentences read, together with how little an oracle bound on retrieval closes it.

Two quantities are measured against the same budget. The first is whether the evidence has been read at all. The second is whether the claim is labelled correctly. They are usually treated as one and they move very differently.

## 2. Related work

Claim verification against an encyclopaedia was set as the FEVER (Fact Extraction and VERification) shared task [1], [2], [4]. Its baseline pipeline follows the open-domain reading model of Chen et al. [5], in which a sparse index narrows the corpus before a reader works, a description that full-text inspection of [1] confirms directly. UKP-Athene [6] is a later shared-task pipeline that instead implements its own entity-linking approach for document retrieval, followed by sentence selection and entailment. Churina et al. [7] improve evidence retrieval for claim verification with question enrichment, but their system is built and evaluated on AVeriTeC, a real-world benchmark their own introduction distinguishes from FEVER's synthetic Wikipedia-derived claims, so this is retrieval-quality work from the same workshop lineage rather than an extension of the FEVER or UKP-Athene pipeline itself.

The shared task's own primary metric already conditions one of these quantities on the other. The FEVER score is label accuracy counted only where at least one complete set of annotator evidence has been retrieved, with not-settled claims exempt from the evidence requirement. A correctly labelled claim supported by no evidence or by only part of a set scores nothing [4]. Reporting that single number is not the same as pricing the two quantities separately against a swept budget. The FEVER score answers how often a system is right with the evidence it has at one operating point; the question here is how much reading each quantity costs and how differently the two respond to paying more. The same source records that predicting not settled for every claim reaches the chance level on that metric, which bears directly on the per-label pattern reported in Section 6.

Withholding a verdict when evidence is insufficient is the reject option Chow [8] analysed, which carries the name selective classification in the modern deep-learning literature, where a classifier declines to predict so as to hold risk to a target level at some coverage [9]. That connection is an analogy rather than an instance. A classifier with a reject option withholds a decision it was asked to make, trading coverage against risk; the verifier here never withholds, because not settled is one of three classes it must choose between and is scored like any other label. Not settled is a property the corpus asserts about a claim rather than an abstention the verifier elects, so nothing below measures a risk-coverage trade-off. Automated fact-checking has also been surveyed directly. Guo, Schlichtkrull and Vlachos [10] cover the pipeline from claim detection through evidence retrieval to verdict prediction and explanation; Kotonya and Toni [11] survey the field with an explicit focus on whether a system's output lets a reader see why a verdict was reached.

Two works anticipate part of this diagnosis and are distinguished here. Akhtar, Schlichtkrull and Vlachos's Ev2R [3] argues that standard evaluation conflates evidence quality with verdict correctness, the same conflation Table 1 separates below. Their remedy is a single combined metric scoring evidence alignment and verdict support together, where this paper instead holds the two quantities apart and prices the gap between them in sentences read at an explicit swept budget, a comparison drawn here from Ev2R's abstract rather than its full method. Pandian's single-author, unreviewed abstention study [12] reports a negative result of the same shape on FEVER-adjacent data: an evidence-conflict-aware abstention rule does not beat

a confidence-margin baseline at matched coverage on a 1,200-claim FEVER sample plus a 209-claim SciFact holdout. Its mechanism is an abstention rule against a confidence baseline, not the oracle promotion of annotator sentences over the 9,999 held-out claims measured below.

The same pattern, a nominal increase in available material not translating proportionally into better use of it, has been shown for long-context language models choosing among passages rather than claim verifiers choosing among retrieved sentences [13]. That a verifier can look accurate without using evidence content discriminatively was shown directly by Schuster, Fisch and Barzilay [14], who found FEVER-trained verifiers insensitive to exactly the kind of fine-grained evidence difference that should flip a verdict. The feature-only verifier used throughout this paper, given no sentence text at all, corroborates and sharpens that finding under a controlled, budget-indexed design rather than establishing it here for the first time. This study measures the cost of retrieval in sentences read against the decision it buys, treating withholding as an outcome rather than a failure.

## 3. Corpus, pool and cost

The claim set is balanced: NOT ENOUGH INFO 6,666, SUPPORTS 6,666, REFUTES 6,666. It is the shared-task development split; the data release [2] names the corresponding file shared_task_dev.jsonl, retrieved here as fever_dev.jsonl with its digest recorded in the study artifacts. The dataset labels NOT ENOUGH INFO, SUPPORTS and REFUTES are written here as not settled, supported and refuted. A claim labelled not settled has no annotator sentence by construction, so evidence recall is defined over the 13,332 claims that can be settled. The verifier is fitted on a random half of the claim set and every accuracy and per-label recall reported is measured on the held-out second half of 9,999 claims.

The retrieval pool is every page the claim set cites plus a sample of the rest: 2,892 cited pages and 119,599 sampled distractors, in all 122,491 pages and 572,664 sentences, drawn from 5,416,537 pages scanned. A smaller pool makes retrieval easier than the full snapshot, so every cost reported here is a lower bound.

Retrieval is a TF-IDF-weighted sentence index [15] over the pool, scored by cosine similarity between claim and sentence vectors; cost is the number of retrieved sentences the verifier is given. The verifier itself sees 10 summary statistics of those sentences: the top and mean score; the gap and decay of the score list; and the maximum, mean and spread of token coverage between the claim and each sentence. It also sees the number of sentences covering at least half of the tokens of the claim, the token count of the claim and the number of distinct pages the retrieved set spans. No feature represents the meaning of a sentence and that is the measurement rather than an oversight. The verifier is a gradient-boosted tree classifier [16], the histogram-based gradient boosting classifier of scikit-learn [17], over those statistics, trained on a random half of the claims and scored on the other 9,999 claims.

## 4. What the budget buys

A hundredfold increase in reading takes evidence recall from 0.5892 to 0.9863 and the decision from 0.4001 to 0.4281 (Fig. 1). Under the complete-set definition the recall curve runs lower at every budget, from 0.5390 to 0.9453, so even after a hundred sentences about one settleable claim in eighteen still has no complete annotator set retrieved. The accuracy difference is paired on the same claims: 0.0280, 95 per cent interval 0.0162 to 0.0396, McNemar p below 0.0001 on 3,640 discordant claims. The first is nearly solved by twenty sentences; the second has moved by under three points across the whole range. Macro F1 tracks accuracy closely at every budget, differing by at most 0.0009, consistent with the three labels being balanced in the claim set.

The accuracy gained by reading more does not come from the evidence arriving. Splitting the held-out claims by whether a complete annotator set was retrieved separates three groups between one sentence and a hundred. Accuracy rose by 0.0418 for the 2,772 claims whose set arrived in that range (interval 0.0198 to 0.0642) and by 0.0505 for the 3,602 whose set had still not arrived at a hundred sentences (0.0303 to 0.0711), while for the 3,625 that already had a complete set at one sentence it did not move at all: -0.0050, interval -0.0240 to 0.0141. The group that never receives its evidence improves at least as much as the group that

Table 1. Evidence recall and verification against the number of sentences read. Evidence recall is given both ways: at least one annotator sentence retrieved, which is the curve plotted in Fig. 1, beside a complete annotator set retrieved, which is what the shared task's own score requires. Right page read counts the claims for which at least one page an annotator cited is among the pages the retrieved sentences come from. Chance for three balanced labels is 0.3333. Accuracy comparisons between two budgets are paired, because both are scored on the same 9,999 held-out claims; the paired intervals are in the artifact repository named in the declarations.

| Sentences read | Annotator sentence read | Complete set read | Right page read | Labelled correctly | Macro F1 |
|---|---|---|---|---|---|
| 1 | 0.5892 | 0.5390 | 0.8372 | 0.4001 | 0.4008 |
| 2 | 0.7221 | 0.6669 | 0.9122 | 0.4117 | 0.4124 |
| 5 | 0.8489 | 0.7895 | 0.9609 | 0.4080 | 0.4089 |
| 10 | 0.9089 | 0.8521 | 0.9791 | 0.4223 | 0.4229 |
| 20 | 0.9496 | 0.8977 | 0.9876 | 0.4229 | 0.4237 |
| 50 | 0.9761 | 0.9302 | 0.9939 | 0.4183 | 0.4190 |
| 100 | 0.9863 | 0.9453 | 0.9965 | 0.4281 | 0.4288 |

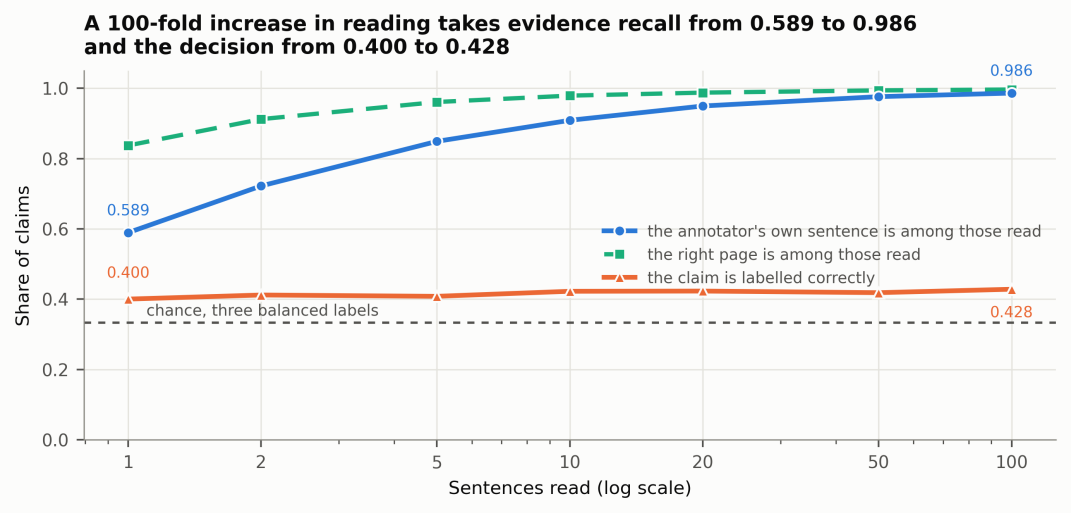



Fig. 1. Evidence recall and verification accuracy against sentences read. The recall curve plotted here is the at-least-one-sentence measure; the complete-set measure is the second column of Table 1.

does, so what a larger budget changes for this verifier is the retrieval statistics themselves rather than the presence of the right sentence. The oracle arm below reaches the same conclusion by a different route.

## 5. What better retrieval would buy

If the decision is limited by the evidence it has, giving it better evidence should help. The oracle arm tests that directly: for every claim that can be settled, the annotator's own sentences are promoted to the top of the retrieved list. No analyst could do this, so it is an upper bound rather than a system.

The mean difference is +0.0072 in accuracy (Table 2) and the paired comparison places all of it at the smallest budgets: +0.0253 at one sentence, +0.0168 at two and +0.0130 at five, then, on this split, nothing measurable from ten sentences on, where every interval in Table 2 straddles zero. Repeating the study over 20 seeds, which resample both the split and the classifier, keeps the shape while softening its edges: the gain at one sentence averages +0.0197 over those seeds against the +0.0253 on this split, it is positive on 20 of 20, the value at five sentences is the largest of the twenty, while across seeds the gain declines smoothly rather than vanishing at ten. The artifact repository carries that comparison. Handing the verifier exactly the evidence a human annotator used changes almost nothing once ten sentences have been read, which locates the remaining gap in using the evidence rather than in finding it. Accuracy in that arm is

Table 2. Accuracy (share of claims labelled correctly) with retrieved evidence against the annotator's own sentences placed first; the difference is the annotator-first column minus the retrieved column. Both arms are scored on the same 9,999 held-out claims, so the difference is paired and the interval is a paired bootstrap over claims.

| Sentences read | Retrieved evidence | Annotator sentences first | Difference | 95% interval |
|---|---|---|---|---|
| 1 | 0.4001 | 0.4254 | +0.0253 | +0.0162 to +0.0344 |
| 2 | 0.4117 | 0.4285 | +0.0168 | +0.0075 to +0.0261 |
| 5 | 0.4080 | 0.4210 | +0.0130 | +0.0045 to +0.0215 |
| 10 | 0.4223 | 0.4207 | -0.0016 | -0.0101 to +0.0070 |
| 20 | 0.4229 | 0.4197 | -0.0032 | -0.0115 to +0.0050 |
| 50 | 0.4183 | 0.4162 | -0.0021 | -0.0096 to +0.0054 |
| 100 | 0.4281 | 0.4300 | +0.0019 | -0.0056 to +0.0094 |

Table 3. Recall and precision by label against sentences read.

| Sentences read | Supports | Refutes | Not settled | Supports prec. | Refutes prec. | Not settled prec. |
|---|---|---|---|---|---|---|
| 1 | 0.3926 | 0.3940 | 0.4145 | 0.4627 | 0.3761 | 0.3735 |
| 2 | 0.4032 | 0.3922 | 0.4410 | 0.4790 | 0.3905 | 0.3797 |
| 5 | 0.4035 | 0.3883 | 0.4333 | 0.4804 | 0.3747 | 0.3833 |
| 10 | 0.4026 | 0.4059 | 0.4602 | 0.4919 | 0.3985 | 0.3928 |
| 20 | 0.4267 | 0.3919 | 0.4512 | 0.4959 | 0.3973 | 0.3888 |
| 50 | 0.4155 | 0.3880 | 0.4527 | 0.4886 | 0.3911 | 0.3887 |
| 100 | 0.4223 | 0.3985 | 0.4651 | 0.5002 | 0.3974 | 0.4007 |

not monotonic in the budget, falling from 0.4285 at two sentences to 0.4162 at fifty. Where that minimum falls is resampling variation, since over 20 seeds it moves to six different budgets. One part of the shape is not: the arm rises from one sentence to two on every seed and falls from two to five on all but one of them; the arm a system can run rises over that step. Placing the annotator sentences first therefore helps most when almost nothing is read, because reading past two sentences adds only non-gold material. The whole one-to-a-hundred difference is +0.0046 on this split, with an interval of -0.0073 to +0.0163 that does not separate it from zero, though it is positive on 16 of 20 seeds, a sign test over those seeds giving 0.0118. The bound is a bound on this verifier and this intervention. It says what promoting annotator sentences is worth to a classifier reading retrieval statistics, not what better retrieval would be worth to a system that read the sentences as language. For a system of this kind, spending on retrieval is spending on the part that is already done.

## 6. Which decision the budget helps

Deciding that a claim is not settled has the highest recall at every budget, reaching 0.4651 at a hundred sentences against 0.4223 for supported and 0.3985 for refuted (Table 3 and Fig. 2). Its recall also rises the most with the budget, by 0.0506 from one sentence to a hundred against 0.0297 for supported and 0.0045 for refuted. A reading consistent with that pattern is that absence of evidence shows up in retrieval statistics as weak scores and poor token coverage while the difference between support and refutation does not, because both look like a strong match. This study does not test that mechanism at the feature level, only through the per-label recall reported here.

For a system whose obligation is not to assert what it cannot support, that is the useful direction. The decision it is required to get right is the one these statistics support best.

## 7. The language model arm

The specification for this study asks for an executed language model producing evidence-linked claims, with retained prompts and raw outputs; this run retained the raw outputs and a recorded prompt template

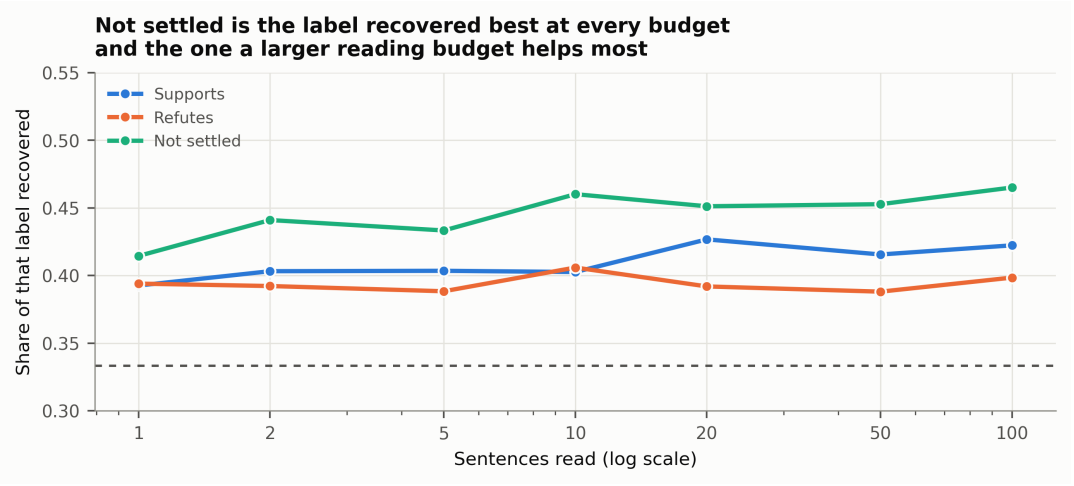



Fig. 2. Which kind of claim a larger reading budget helps. Recall by label is plotted here; precision by label is reported beside it in Table 3.

rather than the literal per-claim prompts, which vary with the retrieved evidence at each budget. An mlx-community/SmolLM3-3B-4bit model [18] was run under greedy decoding with a single label requested per completion, on a stratified 198-claim subsample of the held-out half at budgets 1, 5, 20, 100. Its accuracy over all claims is 0.4697 at budget 1 and 0.3485 at budget 100 against 0.4001 and 0.4281 for the feature verifier at those same two budgets: the language model leads at the smaller budgets and the feature verifier leads at the larger ones. Those two sides were measured on different samples. Scoring the feature verifier on exactly the 198 claims the language model answered makes the comparison matched: it reaches 0.3788 at one sentence and 0.3737 at a hundred against the language model's 0.4697 and 0.3485. Paired on the same claims, none of the four differences is separable from zero; the largest, +0.0909 to the language model at one sentence, has an interval of +0.0000 to +0.1818 with McNemar p of 0.0662. The direction is consistent across budgets, the language model ahead while the budget is small and behind once it is large, but 198 claims do not settle any one comparison. This arm bounds the question rather than answering it.

The run and its scope are recorded in the manifest rather than smoothed over: a single model at a single quantisation with one prompt, a small subsample rather than the full held-out half, no control for whether the model saw this corpus during training and no separately retained record of the exact per-claim prompt text. What the study measures beyond this arm is the part a language model sits on top of: how much evidence a budget buys and how little of the decision that evidence alone settles.

## 8. Reproduction detail

Every value in Sections 3 to 6 comes from one executed run whose identifier is in the manifest with the claim-file hash and the pool sampling record; the retrieval pool, the top 100 retrieved sentences for every claim, evidence recall and verification at seven budgets and the oracle arm are retained with that run. The language model arm in Section 7 comes from a second executed run with its own manifest, its own identifier, retained raw outputs and a recorded prompt template; the exact per-claim prompt text, which varies with the retrieved evidence at each budget, was not separately retained. To obtain the per-claim records the paired comparisons need, both arms were re-run from the same verified input file: Sections 3 to 6 reproduced exactly, while the language model arm reproduced one and a hundred sentences exactly and moved by three claims at five and two at twenty, which is the quantised decoding rather than a change of method. The artifact repository carries both runs, the prompt template verbatim, the paired tables, the per-claim evidence record and the supplementary document itself.

## 9. Discussion and limitations

The first recommendation is to report the budget. An accuracy figure for a verification system is a figure at a reading cost and on this corpus that cost changes evidence recall by forty points while changing accuracy by under three. A reader cannot tell which system is better without both.

The second recommendation follows from the oracle arm. For a verifier working from retrieval statistics, the evidence is already there by twenty sentences and improving the retriever is worth under one accuracy point on average and nothing measurable from ten sentences on. Effort belongs in reading the sentences rather than in finding them. A similarly negative result for evidence-awareness appears in Pandian's abstention study [12], an unreviewed preprint reporting a comparable finding on a far smaller FEVER-adjacent sample through a different mechanism; it does not settle or weaken the bound reported here, which rests on all 9,999 held-out claims rather than 1,200.

Boundaries apply. The language model arm above ran on a subsample at four of the seven budgets rather than the full held-out half, so it bounds rather than settles whether reading the sentences beats reading only their statistics. Matched on the same claims it is ahead at the smaller budgets and behind at the larger ones, by margins none of which the sample settles, so a system that could read the sentences did not simply do better at every budget. The verifier reads summary features rather than language, so the accuracy reported is what a system that cannot read achieves. The pool is 122,491 pages of a snapshot of 5.4 million pages, which makes retrieval easier and makes the costs a lower bound. Cost counts sentences read and not the cost of producing the ranking. The claims are written to be checkable against this snapshot, so the share that cannot be settled is 0.3333 by construction rather than by nature.

## 10. Conclusion

That evidence quality and verdict correctness are conflated in standard reporting is argued elsewhere [3], as is that a verifier can look accurate without using evidence content discriminatively [14]. What this paper adds is a specific, swept-budget measurement of the gap between them together with an oracle bound on how far better retrieval alone could close it. On 19,998 claims against an encyclopaedia snapshot, raising the reading budget from one sentence to a hundred takes evidence recall from 0.5892 to 0.9863 (from 0.5390 to 0.9453 if a complete annotator set is required) while the decision moves from 0.4001 to 0.4281 against a chance of 0.3333. Promoting the annotator's own sentences to the top of the list is worth +0.0253 at one sentence and nothing measurable from ten on, so the gap is not in retrieval. Nor is the gain located where the evidence arrives: claims whose complete evidence set never arrives gain more across that range than claims whose set does. Deciding that a claim is not settled has the highest recall of the three decisions at every budget, which is the decision a system obliged not to assert what it cannot support needs most. Its precision is below precision for supported at every budget and below precision for refuted at five of the seven budgets. Every accuracy reported here is the accuracy of a verifier that reads retrieval statistics rather than sentences, so these conclusions bound what such a system achieves and what better retrieval is worth to it rather than what a system reading the evidence as language would achieve. Reading buys evidence; it does not buy a decision.

## Declarations

Funding. TBD. Competing interests. TBD. Author contributions. TBD. Ethics approval. Not applicable to publicly released, non-personal data; to be confirmed by the authors before submission. Data availability. Every dataset used is public. It is identified in Section 3 with its retrieval location and cryptographic hash. Code availability. The study source, the run manifests, the per-claim records and the supplementary document are at https://github.com/bdata-arindam/evidenceloop-paper. Procedia takes no supplementary file with a camera-ready, so the supplementary document accompanying this paper is in that repository rather than attached. Permissions. TBD. AI-use disclosure. Use of generative AI or AI-assisted technologies in manuscript preparation: AI-assisted tools were used for implementation support, plotting and manuscript

editing. Every experimental output, citation, numerical result and interpretation was checked by the authors against the underlying artifacts. AI tools were never treated as a source of empirical evidence and were not permitted to invent or alter experimental results.

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